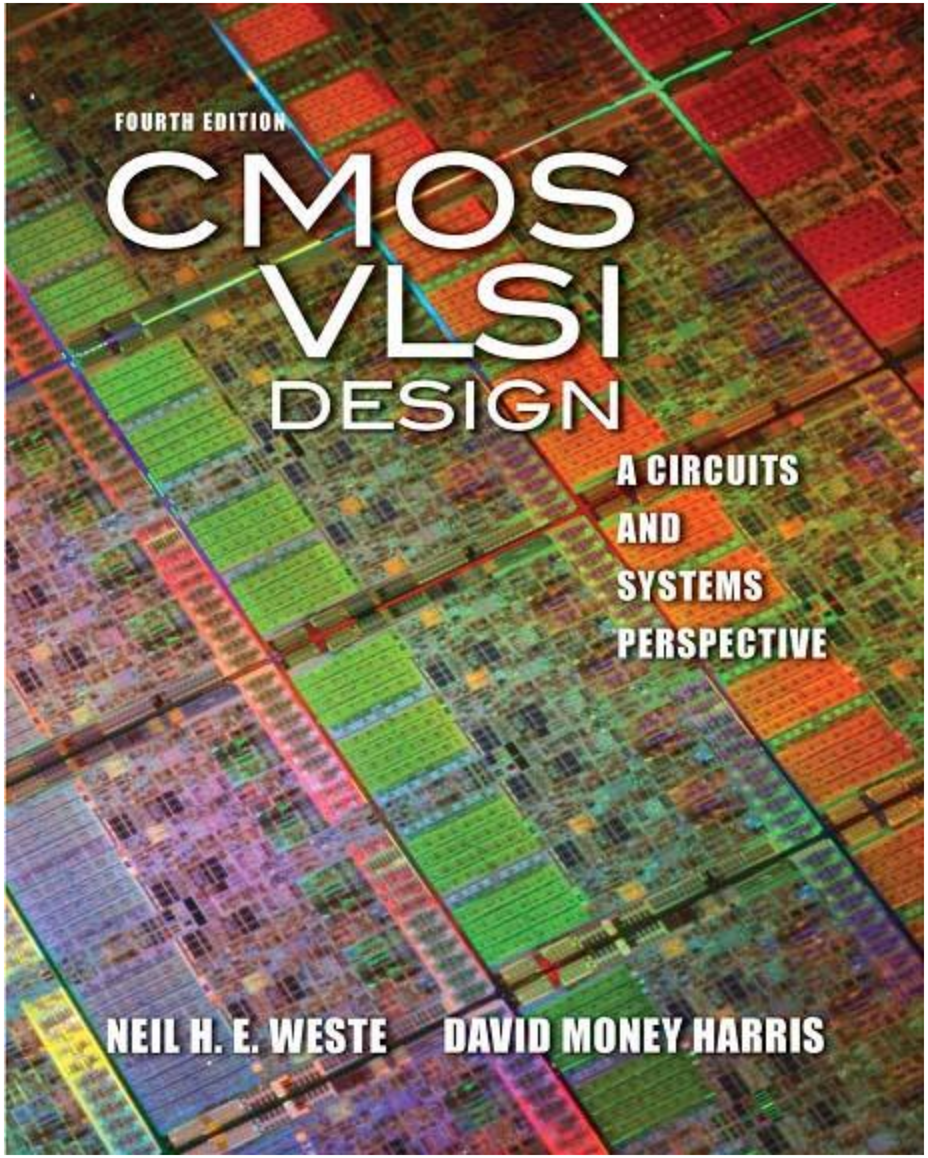


Array Subsystems

Lecture # 4

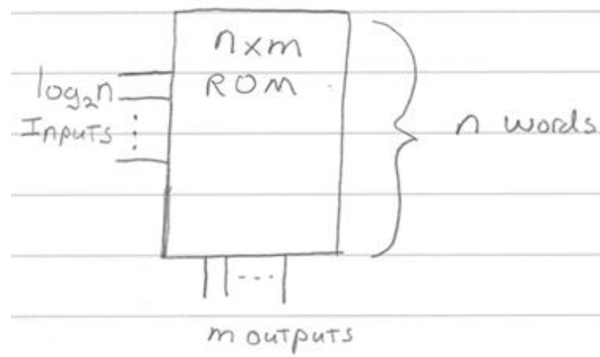
- Boolean Function Implementation with ROM
- Programmable Logic Arrays
  - PLA
  - PAL
  - Boolean Function Implementation with PLA
  - Boolean Function Implementation with PAL

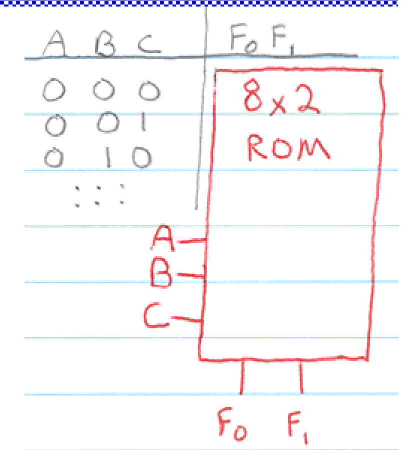
Boolean Function Implementation with ROM

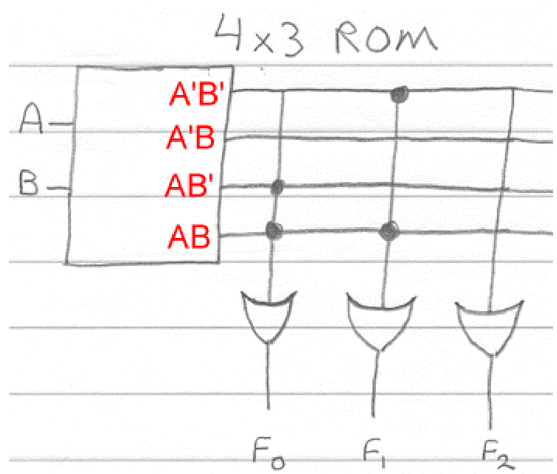
- $F_0 = A$
- $F_1 = A'B' + AB$

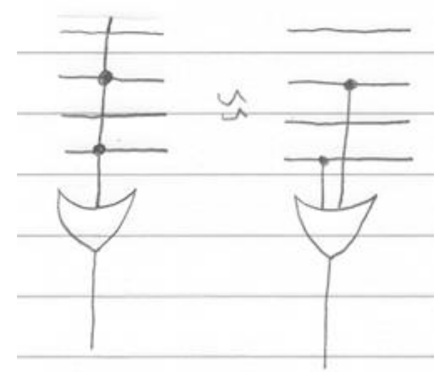
A B	F_0 F_1
0 0	0 1
0 1	0 0
1 0	1 0
1 1	1 1

- $F_0 = AB' + AB$
- $F_1 = A'B' + AB$









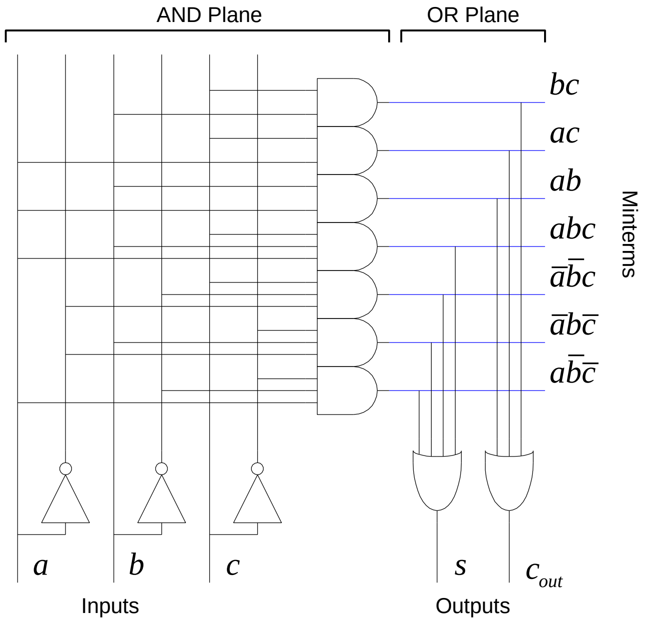
PLAs

- A *Programmable Logic Array* performs any function in sum-of-products form.
- *Literals*: inputs & complements
- *Products / Minterms*: AND of literals
- *Outputs*: OR of Minterms

- Example: Full Adder

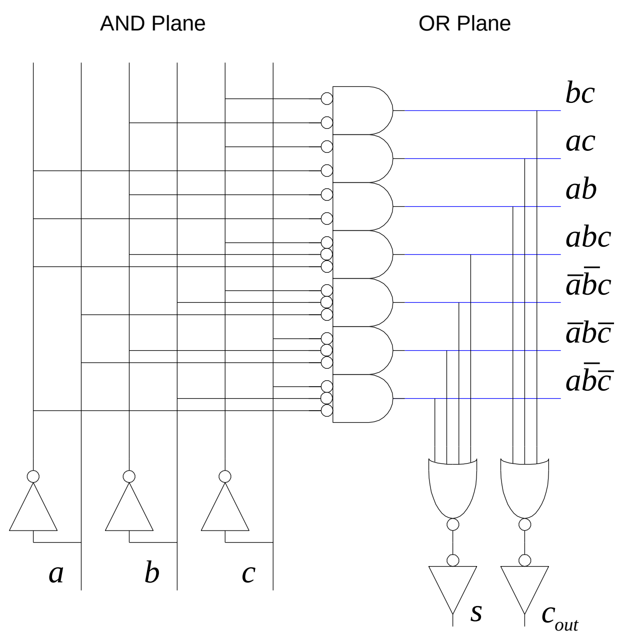
$$s = a\overline{b}\overline{c} + \overline{a}b\overline{c} + \overline{a}\overline{b}c + abc$$

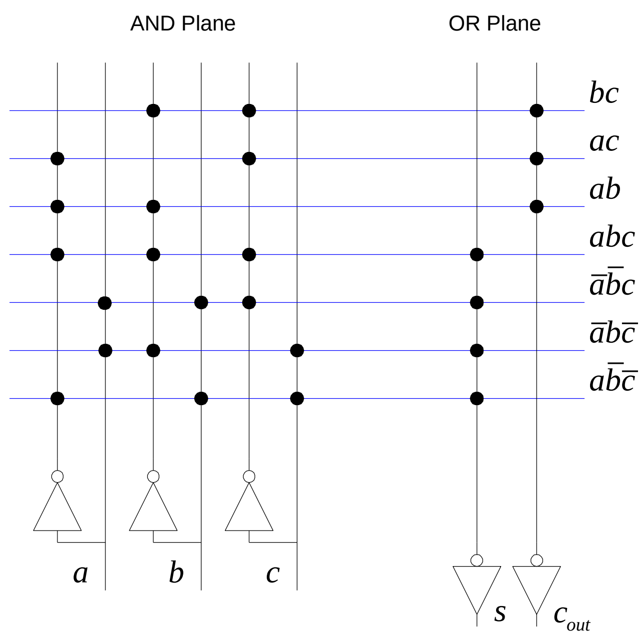
$$c_{\text{out}} = ab + bc + ac$$



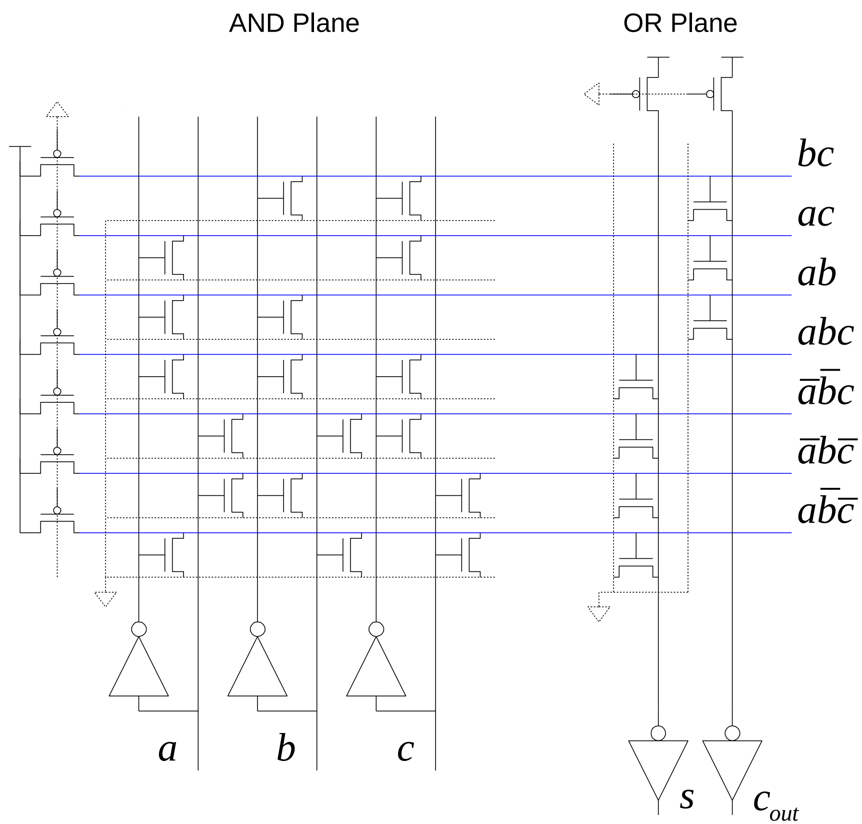
NOR-NOR PLAs

- ANDs and ORs are not very efficient in CMOS
- Dynamic or Pseudo-nMOS NORs are very efficient
- Use DeMorgan's Law to convert to all NORs





PLA Schematic

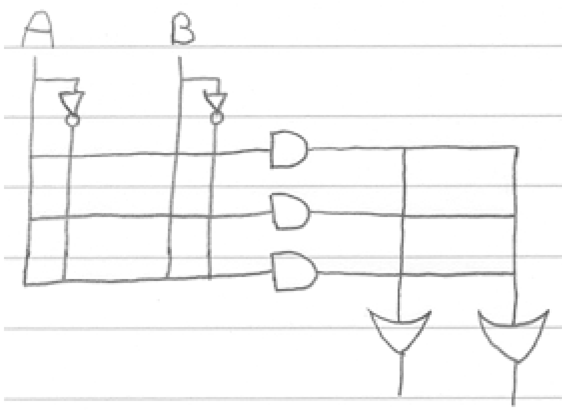


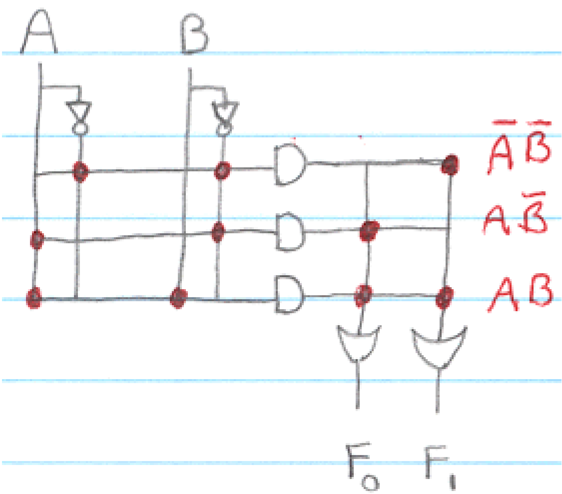
PLAs vs. ROMs

- The OR plane of the PLA is like the ROM array
- The AND plane of the PLA is like the ROM decoder
- PLAs are more flexible than ROMs
  - No need to have 2^n rows for n inputs
  - Only generate the minterms that are needed
  - Take advantage of logic simplification

Boolean Function Implementation with PLA

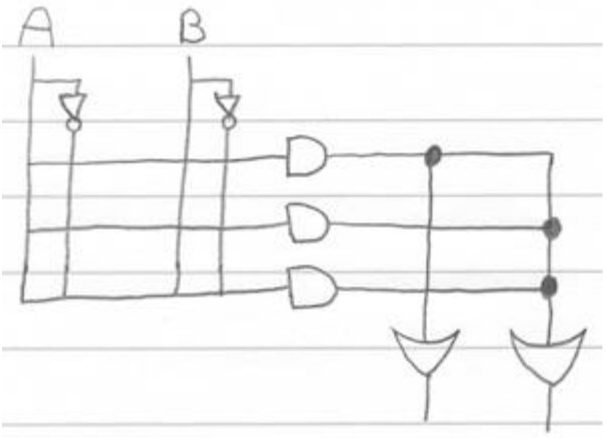
- A PLA is a programmable logic device with a programmable AND array and a programmable OR array.
- $F_0 = A$
- $F_1 = A'B' + AB$

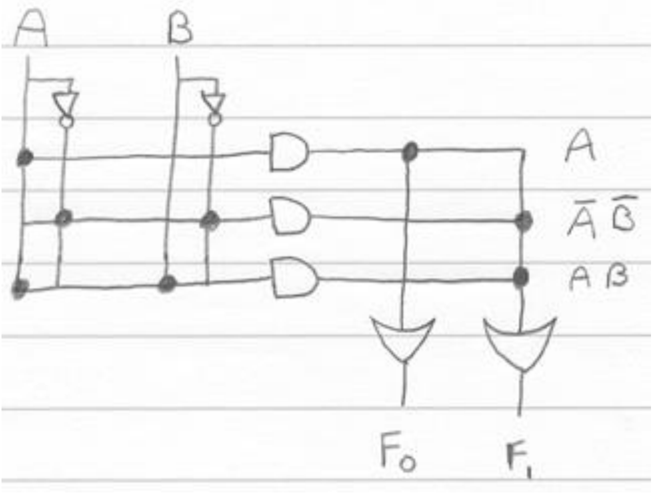




Boolean Function Implementation with PAL

- A PAL is a programmable logic device with a programmable AND array and a fixed OR array.
- $F_0 = A$
- $F_1 = A'B' + AB$





Comparison

- A PLA device has a programmable AND and programmable OR array
- A PAL device has a programmable AND and fixed OR array
- (You could also say that a ROM has a fixed AND and programmable OR array)

Home Work

- Implement the Sum output of Full Adder through ROM, PLA and PAL at transistor level.